

SELMER ELLIPTIC CURVES

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This worksheet studies a single family of plane cubics and uses it to reconstruct, by hand, some of the main results of [Sil09, §§III.1–III.4]: Weierstrass equations, the group law, the Riemann–Roch construction of a Weierstrass model, and a first endomorphism. Sections 1–2 are meant for the problem session; Section 3 may be started there and finished at home; Section 4 is a set of worked Magma examples to read and re-run afterwards, with three short variations to try at the end.

As a rough guide, allow about 12 minutes for §1 and 33 for §2. Within §2, Problem 2.2 is the core and Problem 2.5(e)–(g) the payoff; if time is short, leave Problem 2.5(a)–(d) and Problem 2.6 for later. Section 4 needs no time in the session; it is worked examples, not exercises.

Standing notation. Throughout, k is a field with $\text{char } k \neq 3$, $\bar{k}$ is an algebraic closure of k , and $d \in k^\times$. We write

$$F := X^3 + Y^3 - dZ^3, \quad C_d := \{F = 0\} \subset \mathbb{P}_k^2, \quad O := (1 : -1 : 0),$$

and $\zeta \in \bar{k}$ for a primitive cube root of unity. A pair (C_d, O) of this shape is called a **Selmer elliptic curve**, after [Sel51]; Section 5 explains the name. As usual, X, Y, Z are projective coordinates and lowercase letters are affine coordinates. We write $\overline{C}_d := C_d \times_k \bar{k}$.

1. WARM-UP: A PLANE CUBIC AND A RATIONAL POINT ON IT

Problem 1.1 (It is an elliptic curve).

Recall from [Sil09, §III.3] that an elliptic curve over k is a pair (E, O) with E a smooth, projective, geometrically irreducible curve of genus one over k and $O \in E(k)$ a fixed rational point.

- Check that $O \in C_d(k)$.
- Compute the three partial derivatives of F and deduce that C_d is smooth, using $\text{char } k \neq 3$ and $d \neq 0$. Where is each hypothesis used?
- Suppose instead that $\text{char } k = 3$ (only for this problem). Show that over $\bar{k}$ the curve C_d is a *triple line*.^a
- Show that a smooth plane curve is geometrically irreducible.^b
- Conclude from the degree–genus formula that (C_d, O) is an elliptic curve over k .

^aFrobenius.

^bBézout.

Problem 1.2 (The base point is an inflection point).

- Show that the line $Z = 0$ meets C_d in three *distinct* points of $C_d(\bar{k})$, and write them down. For which k are all three in $C_d(k)$?
- Compute the tangent line L to C_d at O , and show that $L \cap C_d = 3O$. (So O is an **inflection point**, or **flex**.)
- In a Weierstrass model, which line plays the role of L ? What is different here?

- (d) Let $\oplus$ denote the group law on C_d with identity O , defined by the chord–tangent construction of [Sil09, §III.2].^a Using (b), show that for $P, Q, R \in C_d(\bar{k})$,

$$P \oplus Q \oplus R = O \iff P + Q + R \text{ is cut out on } C_d \text{ by a line.}$$

^a[Sil09, Proposition III.3.4] identifies $\oplus$ with addition in $\text{Pic}^0 \bar{C}_d$ via $P \mapsto [P - O]$.

2. THE GROUP LAW

Problem 2.1 (Negation, and 2-torsion).

- (a) Show that $\ominus(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$ for every point $(X_0 : Y_0 : Z_0) \in C_d(\bar{k})$. There are two things to check: that the right-hand side lies on C_d , and that it is the third intersection point of C_d with the line through O and $(X_0 : Y_0 : Z_0)$.
- (b) Deduce that the nontrivial 2-torsion of C_d is cut out by the line $X = Y$, and that $\#C_d[2](\bar{k}) = 4$ when $\text{char } k \neq 2$.
- (c) Take $k = \mathbb{Q}$. For which d does C_d have a nontrivial rational 2-torsion point? Write it down, and give the smallest positive integer d for which this happens.

Part (b) of Problem 2.2 is its heart; parts (d) and (e) are then mechanical. If time is short, take (b) on faith and come back to its proof, but do not skip (c): it is what makes (e) painless. The identity of (b) is the polarization of a cubic form; for a diagonal cubic the trilinear form can be written down by hand.

Problem 2.2 (The trilinear form, and the addition formulas).

For vectors $P = (X_P, Y_P, Z_P)$, Q and R in $\bar{k}^3$, define

$$(2.1) \quad \langle P, Q, R \rangle := X_P X_Q X_R + Y_P Y_Q Y_R - d Z_P Z_Q Z_R.$$

- (a) Check that (2.1) is symmetric in P, Q, R , linear in each argument separately, and satisfies $\langle P, P, P \rangle = F(P)$.
- (b) Prove the *expansion identity*: for scalars s, t ,

$$(2.2) \quad F(sP + tQ) = s^3 \langle P, P, P \rangle + 3s^2 t \langle P, P, Q \rangle + 3st^2 \langle P, Q, Q \rangle + t^3 \langle Q, Q, Q \rangle.$$

Where do the 3's come from?^a

- (c) Prove that $3\langle P, P, Q \rangle = \nabla F(P) \cdot Q$, and deduce that for $P \in C_d(\bar{k})$,

$$\langle P, P, Q \rangle = 0 \iff Q \text{ lies on the tangent line to } C_d \text{ at } P.$$

- (d) Let $P_1 = (X_1 : Y_1 : Z_1)$ and $P_2 = (X_2 : Y_2 : Z_2)$ be distinct points of $C_d(\bar{k})$, and parametrize the line through them as $sP_1 + tP_2$. Using (b), show that the third intersection point is at $(s : t) = (B : -A)$ where $A = \langle P_1, P_1, P_2 \rangle$ and $B = \langle P_1, P_2, P_2 \rangle$, and hence that

$$(2.3) \quad P_1 \oplus P_2 = (BY_1 - AY_2 : BX_1 - AX_2 : BZ_1 - AZ_2).$$

Why can A and B not both vanish?

- (e) Let $P = (X_0 : Y_0 : Z_0) \in C_d(\bar{k})$ and put $Q := (Y_0^2 : -X_0^2 : 0)$. Verify that Q lies on the tangent to C_d at P , and use (b) and (c) to derive

$$(2.4) \quad [2]P = (-Y_0(2X_0^3 + Y_0^3) : X_0(X_0^3 + 2Y_0^3) : Z_0(X_0^3 - Y_0^3)).$$

- (f) Check (2.4) against Problem 2.1: what does it give on the line $X = Y$? (The line $X = 0$ is taken up in Problem 2.5.)

- (g) Where does the whole method break down if $\text{char } k = 3$? Compare your answer with Problem 1.1(c).

^aExpand $\langle sP + tQ, sP + tQ, sP + tQ \rangle$ by trilinearity into eight terms and collect using symmetry.

Problem 2.3 (Arithmetic on C_7).

Let $d = 7$, so that $P := (2 : -1 : 1) \in C_7(\mathbb{Q})$ because $2^3 + (-1)^3 = 7$.

- (a) Compute $[2]P$ using (2.4), and simplify. Verify your answer by exhibiting an identity between cubes of integers.
- (b) Compute $[3]P$ using (2.3) applied to P and $[2]P$, and verify it the same way.
- (c) Compute $[4]P$.
- (d) The three rational points you have found have numerators and denominators of rapidly increasing size. What does this suggest about the order of P in $C_7(\mathbb{Q})$? (You are not asked to prove it; see [Sil09, §VIII.9] or [HS00, Part B].)

Fact 2.4 (The Hessian criterion). Let $C = V(F) \subset \mathbb{P}_k^2$ be a smooth plane curve of degree m over a field of characteristic 0 or > 3 , and let

$$H(F) = \det \left(\frac{\partial^2 F}{\partial X_i \partial X_j} \right)_{0 \leq i, j \leq 2}.$$

Then a point $P \in C(\bar{k})$ is an inflection point if and only if $H(F)(P) = 0$. This is classical, and it genuinely fails in characteristic three; see Problem 2.5(h) below.

In Problem 2.5, parts (a)–(c) are the general theory and (e)–(g) the payoff for our family; if time is short, take (c) on faith and go to (e)–(g). Part (d) is the most satisfying: two completely different counts turn out to agree.

Problem 2.5 (Inflection points and 3-torsion).

- (a) Let $C = V(F)$ be as in Fact 2.4. Show that $H(F)$ is a form of degree $3(m - 2)$, so of degree 3 when $m = 3$. Assuming C is a smooth plane cubic, explain why $C \not\subseteq V(H(F))$, and conclude from Bézout [Har77, §I.7] that C has exactly 9 inflection points *counted with multiplicity*.
- (b) Now forget the Hessian. Let C be a smooth plane cubic over $\bar{k}$ and suppose $O \in C(\bar{k})$ is a flex, so that (C, O) is an elliptic curve with the group law of Problem 1.2(d). Show that

$$P \text{ is a flex} \iff [3]P = O,$$

so that the set of flexes is exactly $C[3](\bar{k})$.

- (c) Show that a smooth plane cubic over $\bar{k}$ always *has* a flex, without using the Hessian.^a Deduce that, with no flex singled out, the set of flexes is a torsor under $\text{Pic}^0(C)[3]$; in particular the number of flexes is $\#C[3](\bar{k})$.
- (d) Combine (b) and (c) with the structure of $C[3]$ in characteristic $\neq 3$ to conclude that a smooth plane cubic has exactly 9 flexes. What does comparing this with (a) tell you about the intersection $C \cap V(H(F))$?
- (e) Specialize to C_d . Compute $H(F)$ for $F = X^3 + Y^3 - dZ^3$ and deduce that, as subsets of $C_d(\bar{k})$,

$$C_d[3](\bar{k}) = (C_d \cap \{XYZ = 0\})(\bar{k}).$$

Write down all nine flexes.

- (f) Check (e) against the group law directly: use (2.4) to show that every point $P = (0 : Y_0 : Z_0) \in C_d(\bar{k})$ satisfies $[2]P = \ominus P$.
- (g) Which of the nine flexes lie in $C_d(k)$? Treat the three lines separately, and say what happens for $k = \mathbb{Q}$.
- (h) (★) In characteristic three, $\#C[3](\bar{k})$ is 3 or 1 according as C is ordinary or supersingular, so a smooth plane cubic has 3 or 1 flexes. How is that compatible with (a)?

^aFor any $P_0 \in C(\bar{k})$ the class $\mathcal{O}_C(1) - 3[P_0]$ has degree 0, and multiplication by 3 is surjective on $\text{Pic}^0 C$.

Problem 2.6 (Complex multiplication, ★).

Let $\alpha(X : Y : Z) = (X : Y : \zeta Z)$, a map defined over $k(\zeta)$.

- (a) Show that α maps C_d to itself and fixes O .
- (b) Quote [Sil09, Theorem III.4.8] to conclude that α is a group homomorphism, hence an automorphism of the elliptic curve (C_d, O) over $k(\zeta)$.
- (c) Show that $\alpha^2 + \alpha + 1 = 0$ in $\text{End}(C_d \times_k k(\zeta))$, the ring of endomorphisms of C_d defined over $k(\zeta)$.^a
- (d) Deduce that $\mathbb{Z}[\zeta] \subseteq \text{End}(C_d \times_k k(\zeta))$, so every Selmer curve has complex multiplication. What does [Sil09, §III.10] then predict for $j(C_d)$ and for $\# \text{Aut}(C_d)$ and $\# \text{Aut}(C_d \times_k k(\zeta))$?

^a $\alpha^3 = 1$, and the endomorphism ring of an elliptic curve has no zero divisors, [Sil09, §III.4].

3. FROM RIEMANN–ROCH TO A WEIERSTRASS EQUATION

This section reproduces, for our family, the proof of [Sil09, Proposition III.3.1]: every elliptic curve is isomorphic to a curve in Weierstrass form. You may use the following two facts without proof.

Fact 3.1 (Bézout). For a nonzero linear form h , the scheme-theoretic intersection $h \cap C_d$ is an effective divisor of degree 3 in $\text{Div } C_d$; over $\bar{k}$ it becomes a sum of three points of $C_d(\bar{k})$ counted with multiplicity. Consequently, since $X + Y$ cuts out $3O$ by Problem 1.2(b),

$$\text{div}\left(\frac{h}{X+Y}\right) = (h \cap C_d) - 3O.$$

Fact 3.2 (Riemann–Roch for genus one). For every $n \geq 1$ we have $\dim_k \mathcal{L}(nO) = n$, where for $D \in \text{Div } C_d$

$$\mathcal{L}(D) = \{f \in k(C_d)^\times : \text{div}(f) \geq -D\} \cup \{0\}, \quad \text{div}(f) \in \text{Div } C_d.$$

This is a space of functions defined *over* k ; since C_d is geometrically integral, $\mathcal{L}(D) \otimes_k \bar{k} = \mathcal{L}(D_{\bar{k}})$, so the dimension is the same whether computed over k or over $\bar{k}$. See [Sil09, §II.5] and the proof of [Sil09, Proposition III.3.1].

Problem 3.3 is meant to be finished at home. Parts (a)–(c) go quickly once you see that Fact 3.1 converts “pole order at O ” into “multiplicity of O in $h \cap C_d$ ”. Part (e) is a two-line computation with a surprising answer, and (h) is worth bringing to the next session.

Problem 3.3 (The Weierstrass model of C_d).

Consider the rational functions in $k(C_d)$

$$w := \frac{Z}{X+Y}, \quad v := \frac{X-Y}{X+Y}.$$

For parts (b)–(f) and (h) assume $\text{char } k \neq 2$. Part (g) replaces v by $X/(X + Y) = (1 + v)/2$, which also makes sense in characteristic 2, and reaches a model valid whenever $\text{char } k \neq 3$.

- (a) Using Problem 1.2(a) and Fact 3.1, compute $\text{div}(w)$. Deduce that w has a pole of order exactly 2 at O and no other pole, so $w \in \mathcal{L}(2O)$. Where are its zeros, and what are they in the language of Problem 2.5?
- (b) Show that the line $X - Y = 0$ does not pass through O . Deduce that v has a pole of order exactly 3 at O , so $v \in \mathcal{L}(3O)$. Where are its zeros?
- (c) Using Fact 3.2, explain why $\{1, w\}$ is a basis of $\mathcal{L}(2O)$ and $\{1, w, v\}$ is a basis of $\mathcal{L}(3O)$.
- (d) The seven functions $1, w, v, w^2, wv, w^3, v^2$ all lie in $\mathcal{L}(6O)$. Why must they satisfy a non-trivial linear relation over k ? List their pole orders at O ; which two collide, and why does that force the relation to involve both?
- (e) Find the relation.^a
- (f) Let $E_1: y_1^2 = x_1^3 - 432d^2$. Show that the relation of (e) says that there is a morphism $\varphi_1: C_d \rightarrow E_1$ with $\varphi_1^*x_1 = 12dw$ and $\varphi_1^*y_1 = 36dv$. Write down φ_1 and its inverse explicitly.
- (g) Where exactly did you use $\text{char } k \neq 2$? Show that there is instead a morphism φ_2 with $\varphi_2^*x_2 = 3dw$ and $\varphi_2^*y_2 = 9dX/(X + Y)$ to

$$(3.1) \quad E_2: y_2^2 - 9dy_2 = x_2^3 - 27d^2,$$

which is a Weierstrass equation in the sense of [Sil09, §III.1] valid whenever $\text{char } k \neq 3$. Which of $a_1, \dots, a_6$ are nonzero?

- (h) (★) The relation in (e) has no w^2 , wv , w or v term, although [Sil09, Proposition III.3.1] allows all four. Explain why, using the automorphisms $\sigma = \ominus$ of Problem 2.1 and α of Problem 2.6.^b

^aWork on the affine chart $X + Y = 1$, where $X = (1 + v)/2$ and $Y = (1 - v)/2$, and expand $X^3 + Y^3$.

^bCompute σ^*w , σ^*v , α^*w , α^*v ; then use (d) to argue that the relation spans a one-dimensional space, so it is a simultaneous eigenvector.

4. WORKED EXAMPLES IN MAGMA

Most of you have never opened **Magma**, so this section is written as worked examples rather than exercises: read them, then re-run them yourself and change things. Every line of output below is reproduced from **Magma** V2.28-8 [BCP97], with whitespace tidied and one long type name elided. Three short variations to try on your own are collected at the end.

Warning 4.1 (Closed points in **Magma**). A **closed point** P of C_d is a point of the scheme C_d ; its residue field $k(P)$ is a finite extension of k , and $\deg P := [k(P) : k]$. The closed points of degree one are the rational points, and a closed point P with $k(P)/k$ separable corresponds to an orbit of $\deg P$ points of $C_d(\bar{k})$ under $\text{Gal}(k^{\text{sep}}/k)$. The divisor group $\text{Div } C_d$ is the free abelian group on the closed points [Poo06, §1.9], whereas [Sil09, §II.3] works with points of $C_d(\bar{k})$ and Galois-invariant divisors on $\bar{C}_d$. **Magma** follows the first convention: **Places** returns closed points, **Divisor** builds elements of $\text{Div } C_d$, and **RiemannRochSpace** returns $\mathcal{L}(D)$ as a k -vector space.

Each example continues the previous one; begin with

```
QQ := Rationals();
P2<X,Y,Z> := ProjectiveSpace(QQ,2);
d := 7;
C := Curve(P2, X^3 + Y^3 - d*Z^3);
O := C![1,-1,0];
```

The notation `P2<x,y,z>` names the coordinates so that we can write equations in them, and `c!` coerces a list of three numbers into a rational point of `C`; typing `0 := (1:-1:0);` does not work.

Example 4.2 (Setting up, and a trap). First, Problem 1.1:

```
IsNonSingular(C); // true
Genus(C);         // 1
0;                // (-1 : 1 : 0)
```

The last line is worth a pause. We asked for $(1 : -1 : 0)$ and got $(-1 : 1 : 0)$: a point of $\mathbb{P}^2(\mathbb{Q})$ is defined only up to scaling, and Magma normalizes the representative it stores, here by -1 .

Now ask for the Weierstrass model. The second return value `mp` is the isomorphism from `C` to `E`.

```
E, mp := EllipticCurve(C, 0);
E;
// Elliptic Curve defined by y^2 - 9/49*y = x^3 - 27/2401 over Rational Field
jInvariant(E); // 0
E eq EllipticCurve([0,0,0,0,-432*d^2]); // false
```

That `false` looks alarming, since Problem 3.3(f) says the answer should be $E_1: y_1^2 = x_1^3 - 432d^2$. It is not a contradiction: `eq` compares Weierstrass *coefficients*, and by [Sil09, Proposition III.3.1(b)] a Weierstrass model is unique only up to the changes of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$. The right question is whether the curves are isomorphic, and against which model:

```
Eours := EllipticCurve([0,0,-9*d,0,-27*d^2]);
Eours; // y^2 - 63*y = x^3 - 1323
bool, iso := IsIsomorphic(Eours, E);
bool; // true
IsomorphismData(iso); // [ 0, 0, 0, 1/7 ]
```

So $r = s = t = 0$ and $u = 1/7 = 1/d$: Magma found exactly the model E_2 of Problem 3.3(g), with x_2 and y_2 divided by d^2 and d^3 . The two maps agree as well,

```
DefiningEquations(mp); // [ 3/7*Z, 9/49*X, X + Y ]
```

against our $(3dZ : 9dX : X + Y)$, the ratios being d^2 and d^3 exactly as $u = 1/d$ demands. The moral: for elliptic curves, `eq` is almost never the question you mean to ask.

We can even watch this happen for all d at once, by working over the function field $\mathbb{Q}(d)$ instead of over $\mathbb{Q}$:

```
Fd<D> := FunctionField(Rationals());
P2d<Xd,Yd,Zd> := ProjectiveSpace(Fd,2);
Cd := Curve(P2d, Xd^3 + Yd^3 - D*Zd^3);
Ed, mpd := EllipticCurve(Cd, Cd![1,-1,0]);
Ed;
// Elliptic Curve defined by y^2 - 9/D^2*y = x^3 - 27/D^4
DefiningEquations(mpd);
// [ 3/D*Zd, 9/D^2*Xd, Xd + Yd ]
```

which is (3.1) rescaled by $u = d$, as an identity in d .

Finally, the predictions of Problem 2.6(d):

```
#AutomorphismGroup(E);           // 2
K<zeta> := CyclotomicField(3);
#AutomorphismGroup(BaseChange(E,K)); // 6
```

Example 4.3 (The addition formulas, checked). A Magma function takes arguments and returns a value; here both are sequences of three coordinates, so that (2.3) and (2.4) can be typed almost verbatim.

```
dbl := function(P)
  x := P[1]; y := P[2]; z := P[3];
  return [-y*(2*x^3+y^3), x*(x^3+2*y^3), z*(x^3-y^3)];
end function;

add := function(P1,P2)
  A := P1[1]^2*P2[1] + P1[2]^2*P2[2] - d*P1[3]^2*P2[3];
  B := P1[1]*P2[1]^2 + P1[2]*P2[2]^2 - d*P1[3]*P2[3]^2;
  return [B*P1[2]-A*P2[2], B*P1[1]-A*P2[1], B*P1[3]-A*P2[3]];
end function;

P := [2,-1,1];
Q := dbl(P);      C!Q;    // (5/3 : 4/3 : 1)
R := dbl(Q);      C!R;    // (-1256/183 : 1265/183 : 1)
S := add(P,Q);    C!S;    // (-17/38 : 73/38 : 1)
```

These are the rational points of Problem 2.3. Coercing through `c!` normalizes them, which is why $(15 : 12 : 9)$ prints as $(5/3 : 4/3 : 1)$. Now transport everything to E through `mp` and compare with Magma's own group law:

```
EP := mp(C!P);
mp(C!Q) eq 2*EP;      // true
mp(C!S) eq 3*EP;      // true
mp(C!R) eq 4*EP;      // true
```

Three `true`s: formulas derived from nothing but (2.1) and a 3×3 determinant agree with a general-purpose implementation.

Example 4.4 (Riemann–Roch, mechanically). Magma can carry out Problem 3.3 without being told the answer. The generators `x` and `y` of `FunctionField(C)` are the affine coordinates X/Z and Y/Z of C_d (not Weierstrass coordinates), so our two functions are built from those.

```
FF<x,y> := FunctionField(C);    // x = X/Z, y = Y/Z
w := 1/(x+y);                  // Z/(X+Y)
v := (x-y)/(x+y);              // (X-Y)/(X+Y)
p10 := Place(0);
Valuation(w, p10);              // -2
Valuation(v, p10);              // -3
```

Those are the pole orders of Problem 3.3(a), (b); a valuation of -2 is a pole of order 2. As in Warning 4.1, `Place(0)` is the closed point O and `Divisor` builds an element of $\text{Div } C_d$, so everything here happens over $\mathbb{Q}$. Next, Fact 3.2 itself:

```

D := Divisor(p10);
for n in [1..6] do print n, Dimension(RiemannRochSpace(n*D)); end for;
// 1 1
// 2 2
// 3 3
// 4 4
// 5 5
// 6 6

```

Now ask for a basis of $\mathcal{L}(2O)$ without suggesting one:

```

V2, m2 := RiemannRochSpace(2*D);
[m2(b) : b in Basis(V2)]; // [ x^2 - y*x + y^2, 1 ]

```

which looks nothing like our w . But $X^3 + Y^3 = (X + Y)(X^2 - XY + Y^2)$, and on C_d the left side is dZ^3 , so

$$\frac{X^2 - XY + Y^2}{Z^2} = \frac{dZ^3}{Z^2(X + Y)} = d \frac{Z}{X + Y} = dw,$$

and indeed

```

(x^2 - x*y + y^2) eq d*w; // true

```

Finally the relation. We list the seven functions of Problem 3.3(d), express each in a basis of $\mathcal{L}(6O)$ (that is what @@ does, applying the inverse of the map mV), and take the kernel of the resulting matrix.

```

V, mV := RiemannRochSpace(6*D);
fns := [FF!1, w, v, w^2, w*v, w^3, v^2];
[-Valuation(f, p10) : f in fns]; // [ 0, 2, 3, 4, 5, 6, 6 ]
M := Matrix([Eltseq(f @@ mV) : f in fns]);
K := Kernel(M);
Dimension(K); // 1
Basis(K); // [ ( 1 0 0 0 0 -28 3) ]

```

The pole orders are 0, 2, 3, 4, 5, 6, 6: seven functions, six available orders, the collision at 6 forcing the relation exactly as in Problem 3.3(d). And the kernel is one-dimensional, spanned by the vector reading

$$1 - 28w^3 + 3v^2 = 0,$$

which is $3v^2 + 1 = 4dw^3$ at $d = 7$. We got it by hand from a factorization; Magma got it from linear algebra in a Riemann–Roch space.

Now try these. Each one is a small edit to Example 4.2 or Example 4.3, and in each case the worksheet already tells you what the answer must be.

- (1) Set $a := 2$. Problem 2.1(c) predicts a rational point of order two. Find it, verify with the group law that it has order two, and compute its image under φ_1 by hand.
- (2) Set $a := 8$. Problem 2.5(g) predicts rational 3-torsion. Find the two rational 3-torsion points other than O , use your ab1 function to check that $[2]P = \ominus P$, and compute $\varphi_1(P)$ by hand.

- (3) Use `IsIsomorphic` to decide whether $C_1 \simeq C_8$ and whether $C_1 \simeq C_2$ over $\mathbb{Q}$. Pass it the *Weierstrass* models: on the plane curves themselves `Magma` answers `For genus 0 or 1 curves the ↪ basefield must be finite`. Reconcile the answers with the cubic-twist criterion of Section 5.

5. WHERE THE NAME COMES FROM

Selmer [Sel51] studied the general diagonal cubic $aX^3 + bY^3 + cZ^3 = 0$, and produced the standard counterexample to the Hasse principle: the curve $3X^3 + 4Y^3 + 5Z^3 = 0$ has real points and $\mathbb{Q}_p$ -points for every prime p , but no rational point. Such a curve is a torsor under its Jacobian [Poo17, §5.12], and that Jacobian is a Selmer curve in our sense:

$$\text{Jac}(aX^3 + bY^3 + cZ^3) \simeq (y^2 = x^3 - 432(abc)^2).$$

Two ways to test this in `Magma`: the curve $X^3 + 2Y^3 + 3Z^3 = 0$ *does* have the rational point $(1 : 1 : -1)$, so it is its own Jacobian and should be isomorphic to C_6 , that is to $y^2 = x^3 - 432 \cdot 6^2$; and for Selmer's own curve one can call `Jacobian(GenusOneModel(3*X^3 + 4*Y^3 + 5*Z^3))` and compare with $y^2 = x^3 - 432 \cdot 60^2$. Both come out `true`.

Finally, by Problem 3.3(f) together with the classification of twists of $y^2 = x^3 + A$ [Sil09, §X.5], for $\text{char } k \neq 2, 3$ the curves C_d and $C_{d'}$ are isomorphic over k exactly when $d/d' \in (k^\times)^3$, so the family is parametrized by $k^\times/(k^\times)^3$. These are the **cubic twists** of $X^3 + Y^3 = Z^3$, and they are the natural first examples for the descent machinery of [Sil09, §X] and [Poo17, Ch. 8].

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